

Fake Job Posting Detection

Using Machine Learning

Final Project Report

1. Abstract

Fake job postings are a big problem today. Scammers post fake job ads online to steal money or personal information from job seekers. It is hard for a normal person to always tell if a job posting is real or fake, so this project uses Machine Learning to do that job automatically.

We took a dataset of job postings and looked at the text in them, such as the title, description, and requirements. This text was cleaned and converted into numbers using a technique called TF-IDF, because computers understand numbers better than words. Three different Machine Learning models were then trained: Logistic Regression, Naive Bayes, and Random Forest. After comparing all three, Random Forest gave the best results, so it was chosen as the final model. Finally, a simple system was built that can take any new job posting and tell whether it looks real or fake.

2. Introduction

These days, most people look for jobs online instead of newspapers. This makes it very easy for scammers to post fake job ads too, since anyone can create a listing on a job website. People who apply to these fake jobs can lose money (through fake registration fees) or even have their personal details misused.

Checking every single job posting by hand is not possible because there are thousands of them every day. This is where Machine Learning helps. A computer can be trained to read the text of a job posting and learn the common patterns found in fake jobs, such as unrealistic salaries, spelling mistakes, or requests for money. Once trained, the model can automatically flag suspicious postings, saving time and protecting job seekers.

3. Objectives

The main goals of this project are:

- To study and understand the fake job postings dataset.
- To clean the text data so it is ready for a Machine Learning model.
- To convert the text into numbers using the TF-IDF technique.
- To train different Machine Learning models on this data.

- To compare the models and pick the one that works best.
- To build a simple system that predicts if a new job posting is real or fake.

4. Dataset

This project uses the Fake Job Postings dataset (also called EMSCAD), which is available on Kaggle. It has about 17,880 job postings, each with details like the job title, company profile, description, requirements, benefits, employment type, required education, and required experience.

Every job posting also has a label called 'fraudulent', which tells us the true answer:

- 0 → the job posting is Legitimate (real)
- 1 → the job posting is Fraudulent (fake)

This label is what the Machine Learning model learns to predict.

5. Methodology

The project was carried out in the following simple steps:

- Loading the dataset and exploring what it contains.
- Checking and handling missing values and duplicate rows.
- Cleaning the text: converting to lowercase, removing numbers, punctuation, and extra spaces.
- Combining the important text columns (title, company profile, description, requirements, benefits) into one single text field.
- Converting the cleaned text into numbers using TF-IDF (a method that gives weight to important words).
- Splitting the data into a training set (to teach the model) and a test set (to check how well it learned).
- Training three models: Logistic Regression, Multinomial Naive Bayes, and Random Forest.
- Evaluating each model using Accuracy, Precision, Recall, F1-score, and a Confusion Matrix.
- Comparing all the models to find the best one.
- Trying an alternative method (CountVectorizer) and tuning the Random Forest settings to see if results improve.
- Building a final prediction system that classifies any new job posting.

6. Results

After training and testing, all three models were compared using their accuracy and other scores. Random Forest performed the best overall among the three models, so it was chosen as the final model for making predictions.

Two extra experiments were also done. First, TF-IDF was compared with a simpler method called CountVectorizer, to see which one represents the text better. Second, the settings (hyperparameters) of the Random Forest model were adjusted, such as the number of trees and tree depth, to see if the results could be improved further. Both experiments showed that the way we represent text and configure the model has a real effect on performance.

Model Comparison (Summary)

Model	Result
Logistic Regression	Good baseline performance
Multinomial Naive Bayes	Fast but less accurate than others
Random Forest	Best overall performance — selected as final model
Tuned Random Forest	Further improved with parameter tuning

7. Prediction System

A simple prediction system was built so that a brand-new job posting (that the model has never seen before) can be checked instantly. The system works in the following order:

- A new job posting is entered as text.
- The text is cleaned in the same way as the training data.
- The cleaned text is converted into numbers using the same TF-IDF method.
- The trained Random Forest model looks at these numbers and makes a prediction.
- The system shows the result: Legitimate or Fraudulent, along with a confidence score.

For example, a normal-sounding job posting for a Software Engineer role was correctly predicted as Legitimate, while a posting promising a very high salary for no experience along with a request for a registration fee was correctly predicted as Fraudulent.

8. Conclusion

This project successfully built a Machine Learning system that can detect fake job postings by reading their text. After cleaning the data and testing three different models, Random Forest gave the best results and was chosen as the final model. The prediction system built at the end can be used to automatically check any new job posting and warn users if it looks suspicious. This kind of tool can help protect job seekers from falling victim to employment scams.

9. Future Scope

This project can be improved further in the future in the following ways:

- Using more advanced NLP models such as BERT or RoBERTa for better text understanding.
- Adding more structured features, like salary range and company verification status.
- Turning this project into a live web application using Flask or Streamlit, so anyone can use it easily.
- Retraining the model regularly with new data, so it keeps working well over time as scam patterns change.